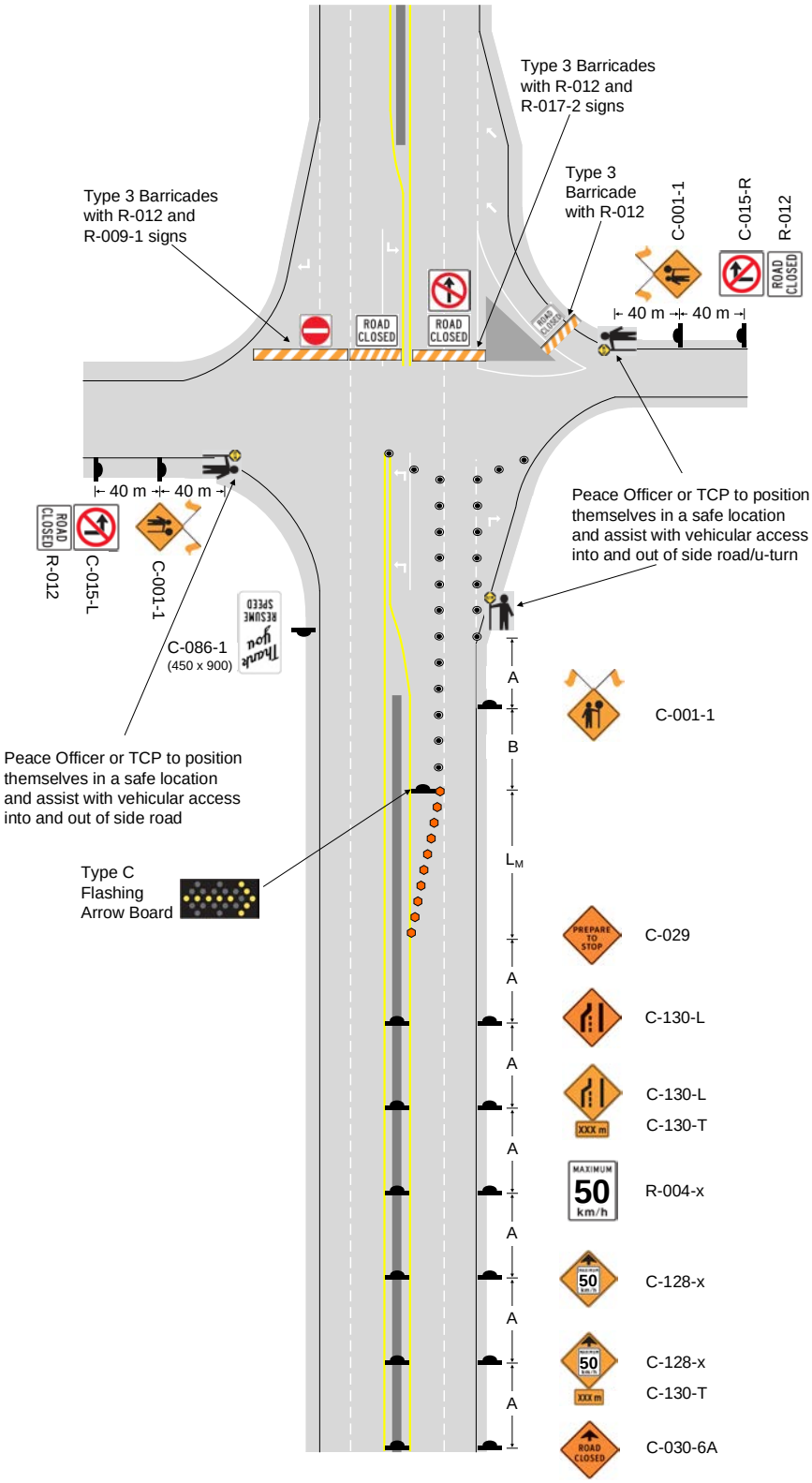
Figure 19.5: Road Closure – Intersection (Signalized or Unsignalized)



19.6 Roadway Closure with Detour (≤ 60 km/h)

Purpose:

This layout shows a typical setup for a local street network where network street continuity is lost due to the closure of a street. Advance planning for detours is key to success, as in some cases using other jurisdictions streets or developing custom signs will be necessary. Getting acceptance from other jurisdictions or getting customized signs developed takes time, so lead time is key. Emergency services (Police, Ambulance, Fire) may need to be contacted depending on the complexity of the closure (network of streets impacted), and overall length (time and distance) of detour.

Standard:

- All detours affecting provincial roadways and highways shall have a plan accepted by the Ministry.
- Before a road is closed to traffic, all necessary detour signs shall be in place along the corresponding detour route.

Guidance:

- Regulatory traffic control devices should be added or modified as needed for the duration of the detour.
- *Figure 19.3: Roadway Closure with Detour (≤ 60 km/h)* shows the general layout of detour signs. Additional detour signs should be erected at all connecting roadways to provide clear guidance.

Options:

- DMS messages may be used to provide enhanced information regarding the closure and detour. Typical messages are shown below:

FIRST AVE CLOSED	BEGINS APRIL 20
ROAD CLOSED	DETOUR MAIN ST

- If the road is opened for some distance beyond the intersection and/or there are significant origin/destination points beyond the intersection such as residences, the Road Closed R-012 and Detour C-005-LR1 signs on barricades may be located at the edge of the traveled way to allow local passage.
- If the road is opened for some distance beyond the intersection and/or there are significant origin/destination points beyond the intersection such as residences, a Road Closed Local Traffic Only C-201 series sign may be used in place of the Road Closed R-012 and Local Traffic Only R-012-T sign assembly on the barricade.
- A Street Name sign may be mounted with the Detour sign.
- Flashing warning lights and/or flags may be used to call attention to the advance warning signs.
- If the road closure impacts a main collector route to a highway, detours may require route shields and the required direction arrows.
- Typical custom signage used in detours include the following examples:

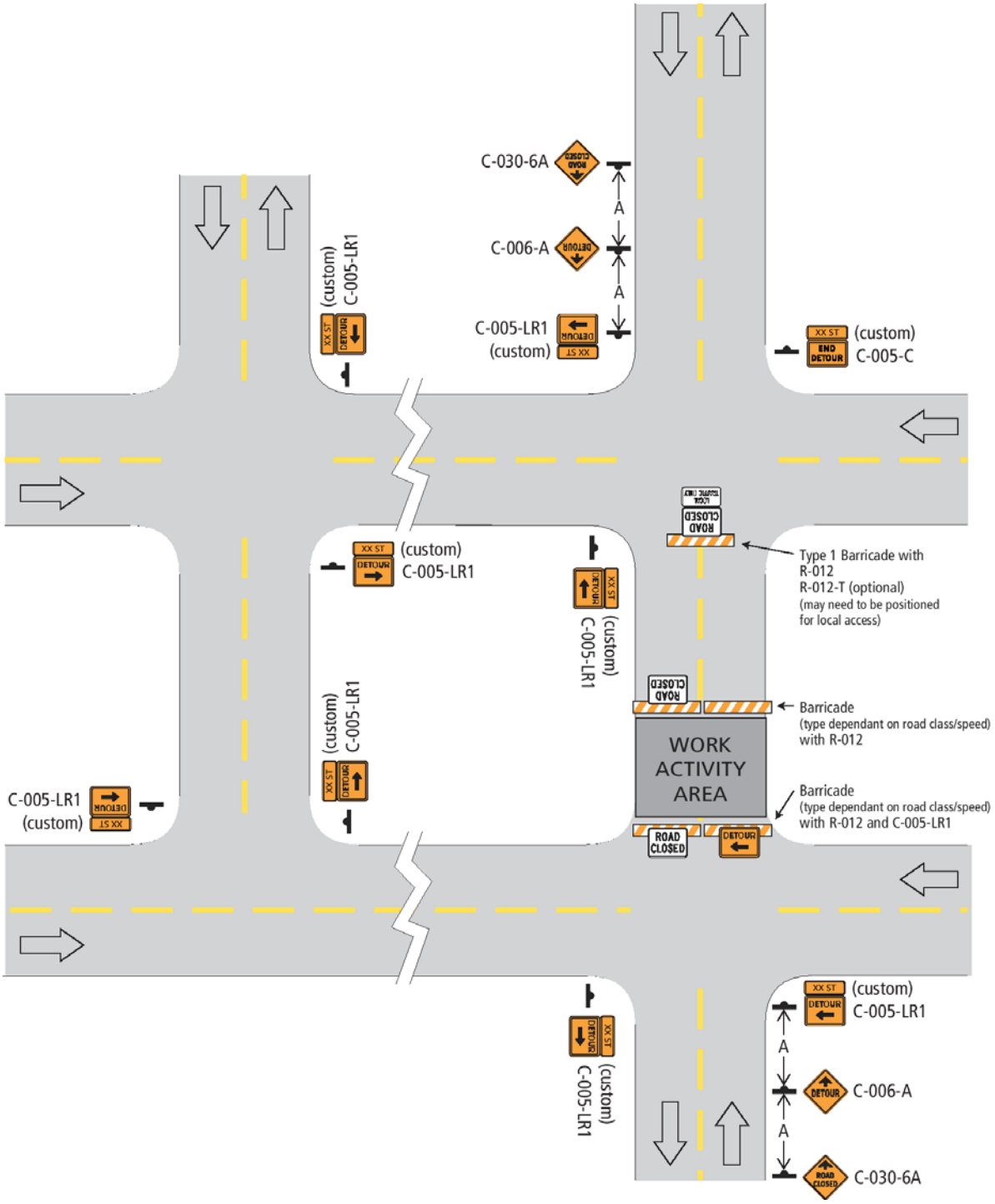






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Figure 19.6: Roadway Closure with Detour (≤ 60 km/h)



19.7 Roadway Closure with Detour (≥ 70 km/h)

Purpose:

This layout shows a typical setup for a long duration highway closure, or main arterial closure where the highway is closed, or if a direction of the highway is closed for more than 3 days. A detailed traffic management plan will be required as advanced planning for detours is key to success, especially in cases where another jurisdiction's roads, or the need to develop custom signs is necessary.

Obtaining acceptance from other jurisdictions or developing custom signs takes time, so lead time is key. Emergency services (Police, Ambulance, Fire) shall be contacted, as incident planning is required. Prior to a planned highway closure, information signs for road users shall be deployed.

Standard:

- All detours affecting provincial roadways and highways shall have a plan accepted by the Ministry.
- Before a road is closed to traffic, all necessary detour signs shall be in place along the corresponding detour route.

Guidance:

- A Traffic Management Plan including the four sub-plans should be developed (Traffic Control Plan, Incident Management Plan, Public Information Plan, and Implementation Plan).
- Advanced information signage regarding the detour should be in place at least 8 days prior to the detour opening, and may require longer notification for lengthy and complex detours.
- DMS messages should be used to provide enhanced information regarding the closure and detour. Typical messages are shown below:

FIRST AVE CLOSED	BEGINS APRIL 20
ROAD CLOSED	DETOUR MAIN ST

- For long duration detours, signage should be placed on permanent structures such as telespar.
- Route shields and directional arrows should be placed to clearly direct all users through the detour route.
- Regulatory traffic control devices should be added or modified as needed for the duration of the detour.
- *Figure 19.4: Roadway Closure with Detour (≥ 70 km/h)* shows the general layout of detour signs. Additional detour signs should be erected at all connecting roadways to provide clear guidance.

Options:

- Placement of detour signage on both sides of the road may be necessary for clearer guidance.
- When adding signs, utilizing the back of detour signs for one direction for the other will minimize the number of sign structures added.
- If the road is opened for some distance beyond the intersection where the detour begins and/or there are significant origin-destination points beyond the intersection such as commercial businesses, the Road Closed R-012 and Detour C-005-LR1 signs on Type 3 barricades may be located at the edge of the traveled roadway to allow local passage.
- If the road is opened for some distance beyond the intersection and/or there are significant origin/destination points beyond the intersection such as residences, a Road Closed Local Traffic Only C-201 series sign may be used in place of the Road Closed R-012 and Local Traffic Only R-012-T sign assembly on the barricade.
- Flashing warning lights and/or flags may be used to call attention to the advance warning signs.
- Typical custom signage used in detours include the following examples:

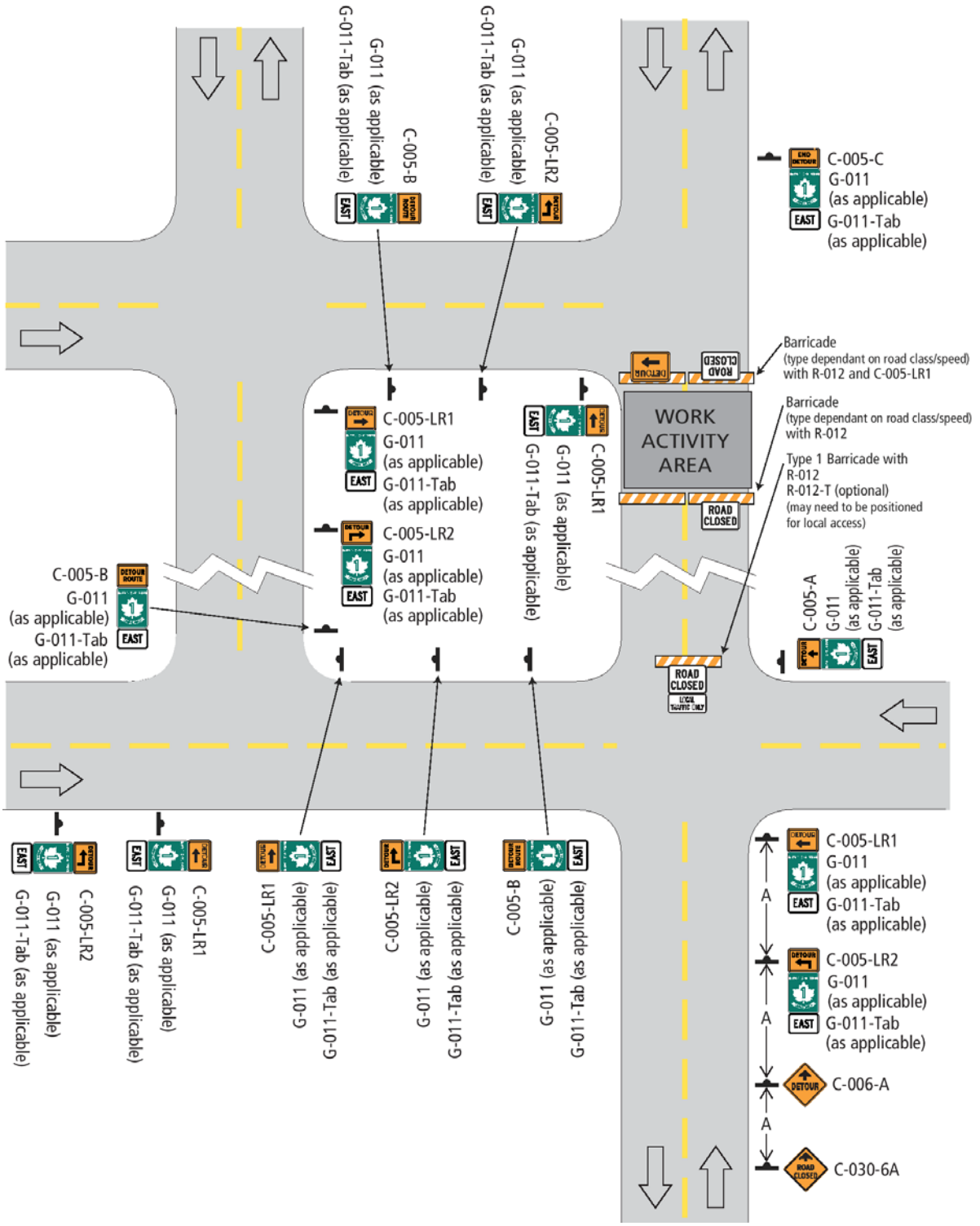






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Figure 19.7: Roadway Closure with Detour (≥ 70 km/h)



19.8 Emergency Response

Purpose:

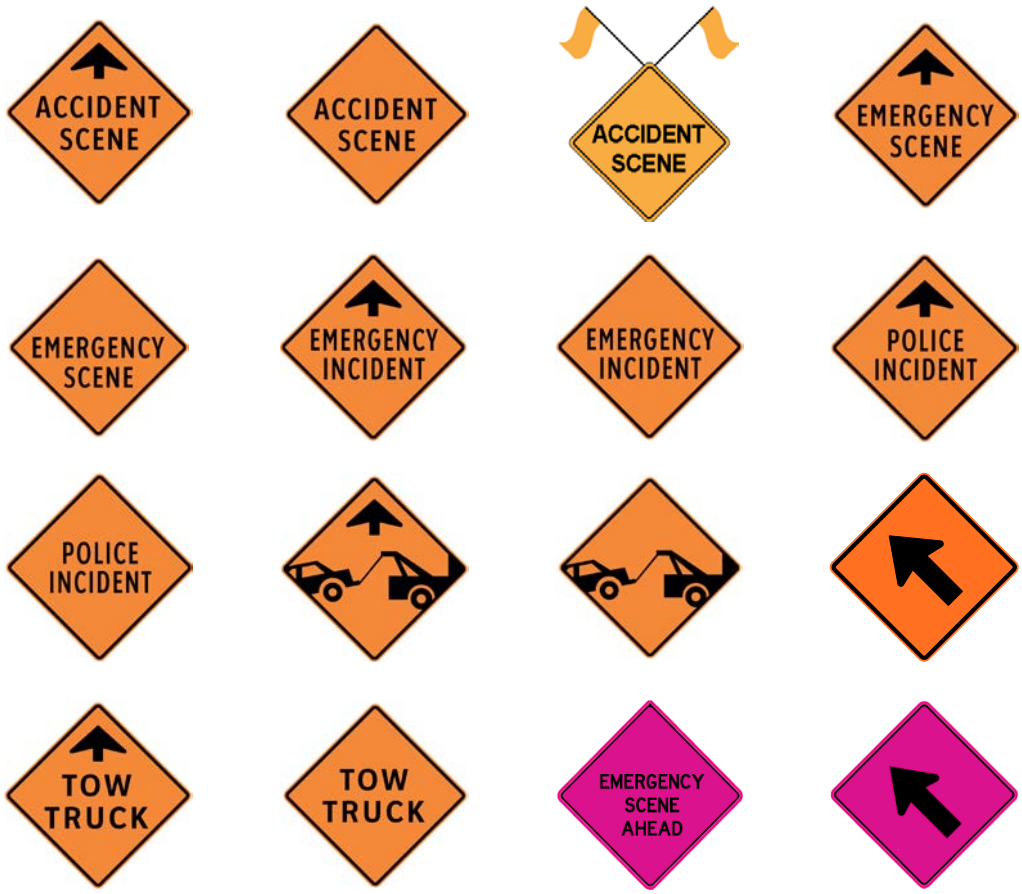
This layout shows a typical setup of signs that may be used by those who attend incidents prior to the arrival of planned and organized traffic management and traffic control. First responders may carry flexible signs that can be set up quickly using portable lightweight spring stands, or other appropriate temporary mounting at an incident site.

The purpose of the signs is to alert drivers that the temporary traffic control is a result of an emergency situation, to expect responders on the roadway, and to proceed with caution as full temporary traffic control may not yet have been established.

C-058 series signs used for emergency response can be either the Ministry's standard retroreflective fluorescent orange or they may be the flexible roll-up retroreflective pink which some organizations have purchased.

Standard:

- Emergency response signs include the following:



Guidance:

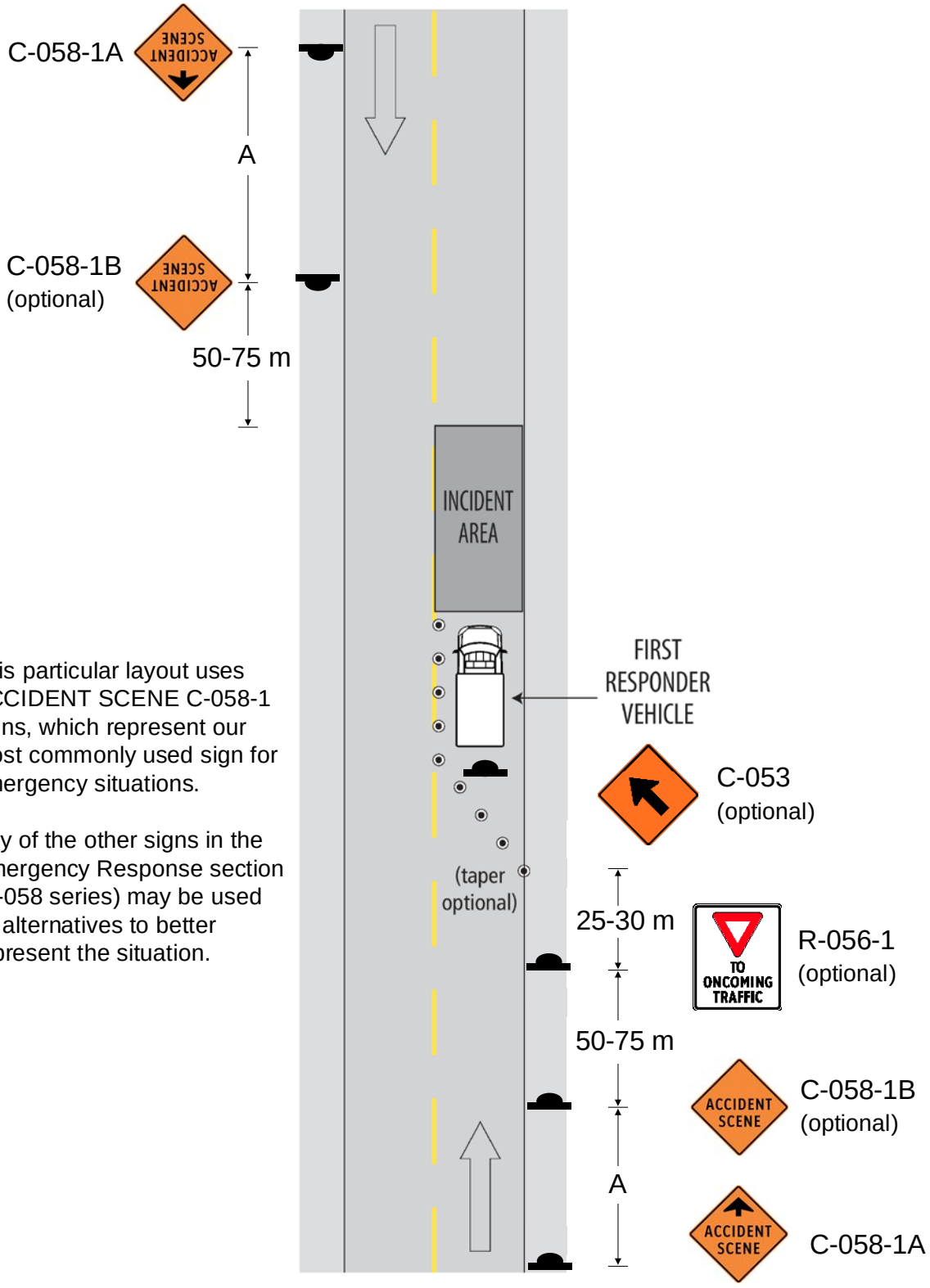
- First responders may carry rigid or flexible roll-up signs, using messages that coincide with their field activity. Default messages include Accident Scene C-058-1 or Emergency Scene C-058-2 series signs.
- Signs placed and positioned by first responders should follow the guidelines established in this Manual.
- Wind and vehicles can blow signs over. Therefore, signs should be placed on the shoulder of the road as far off the travelled portion of the road as reasonably possible.
- An unplanned event such as an emergency leads to a variety of activities taking place at the same time. Those responding to the emergency should only undertake traffic control activities if practical.

Options:

- The Accident Scene Ahead C-058-1A sign (or similar), and the Lane Closure Arrow C-053 sign (if used), should be placed on the shoulder of a roadway by a qualified first responder (fire, enforcement, ambulance, HAZMAT, or recovery) in advance of the incident scene.
- Accident Scene Ahead C-058-1A signs should be placed in both directions so that they will provide enough warning for vehicles to slow down before reaching the incident scene. Placement of an advance warning sign for situations near a corner, hill, or other reduced visibility situations, may require the placement to be adjusted.
- The Accident Scene C-058-1B sign (or similar) may be used to provide additional warning for vehicles to slow down before reaching the incident scene.
- Where the road user is required to navigate around an incident, the Lane Closure Arrow C-053 sign should be placed. If no cones are utilized, the directional arrow sign should be placed in a location which provides enough separation for road users to navigate around the incident.
- If the Lane Closure Arrow C-053 sign is used, it should be placed sufficiently far back from the incident to allow motorists to both see beyond the incident, and easily manoeuvre around the incident.
- If the Lane Closure Arrow C-053 sign is used, it should be located to close off the approach to the incident scene. For example, if the incident is contained within a travel lane, it should be placed at the midpoint of the lane, or towards the centre line or lane line.
- If the incident is blocking a travelled lane on a two-lane two-way roadway, the Lane Closure Arrow C-053 sign requires the addition of the Yield to Oncoming Traffic R-056-1 sign, as the Lane Closure Arrow C-053 sign would otherwise be directing traffic into the opposing lane.

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Figure 19.8: Emergency Response



This particular layout uses ACCIDENT SCENE C-058-1 signs, which represent our most commonly used sign for emergency situations.

Any of the other signs in the Emergency Response section (C-058 series) may be used as alternatives to better represent the situation.

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Appendix A: Glossary

Contents

A.1 Terms A-1

A.2 Acronyms[1] A-15

[1] For sign acronyms and abbreviations, see *Section 4.2: Traffic Signs* and *Section 4.3: Dynamic Message Signs.*

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Appendix A: Glossary

A.1 Terms

active work	When workers and equipment are present on the construction site.
barricade	A frangible device that is relatively forgiving when struck by an errant vehicle. It is normally placed at or nearly at right angles to approaching traffic to provide visual identification of hazardous locations and to delineate travel paths.
barrier	A device typically filled with water, or concrete, and designed to contain and deflect errant vehicles at a small angle, thereby preventing them from entering a closed or hazardous area. It is normally placed parallel to or nearly parallel to approaching traffic.
bicycle lane	A lane designated for bicycle use, and may be marked with both a diamond and a bicycle symbol.
brief-duration work	Work that is generally planned in nature, but for which the extent of the work required or the location may not be fully known. The total time to complete brief-duration work is less than 15 minutes.
buffer space	The longitudinal distance which provides a margin of safety for both the driver and the workers. It is important that the buffer space be free of equipment, workers, material and vehicles. A buffer vehicle with a crash attenuator may be located within the buffer space if there are space constraints. The buffer space is measured is from the end of the taper to the work activity area unless there is a buffer vehicle, in which case it is measured to the back of the buffer vehicle. See also *Section 6.2.4: Work Zone Components – Buffer Space.*
buffer vehicle	A vehicle positioned in the buffer space in advance of a work activity area to enhance worker safety. It is usually stationary, and shall be equipped with a 360-degree flashing yellow light and 4-way flashers, or a flashing arrow board (FAB)
changeable message sign	See “dynamic message sign” in this *Glossary*.

chase vehicle	A separate traffic control vehicle used in a rolling slowdown operation, following the slowest or last public vehicle ahead of the blockade (see *Section 10.6: Rolling Slowdown*).
clear zone	From the Ministry's BC Supplement to the Transportation Association of Canada's Geometric Design Guide: The total roadside border area—starting at the edge of the travel lane—that is available for errant vehicles. A clear zone may consist of a shoulder, a recoverable slope, a non-recoverable slope, and/or a clear run-out area.
construction speed limit	The speed limit (unchanged or reduced) in the work zone during construction activities or other temporary/special events.
Construction Speed Zone	A lowered legal speed zone authorized by the Road Authority, normally through a long-duration work zone.
continuously slow-moving work	Work that is continuously moving such that the use of normal traffic control procedures is impracticable. This can include stops of up to 15 minutes. Examples include hydro-seeding, spraying for dust control, grading, mowing, brushing, flushing, striping, and sweeping.
conventional pavement marking	A form of pavement marking in which the paint dries slowly (drying time is 5 minutes or more), and from which paint could be tracked into travel lanes by vehicles driving over the marking. It requires advance warning signage and protection of the painted area by using signs, cones, barricades, and other devices until the paint is dry. See also "quick-dry pavement marking" in this *Glossary* and *Section 14.1: General Information* on pavement marking.
detour route	A travel route that takes traffic off the normal route and uses existing roadways or new temporary roadways to guide traffic around a work zone, identified by appropriate detour signs. At least one week prior to closing a roadway and opening a detour, it is advisable to erect "Closing Notice" signs at strategically selected locations.
directional dividing line	A yellow line that separates traffic traveling in opposite directions. Directional dividing lines are also used to mark the left edge line of divided highways and one-way roadways, including portions of highway ramps, and to mark both sides of two-way left-turn lanes.

divided roadway

Roadway separated by median barrier (concrete, cable and W-beam), depressed median (≥ 4 metres wide), raised channelization with curb (≥ 2 metres wide), or landscaping with curb.

downstream

Like the flow of a river, a location away from a present location, in the direction of vehicular travel.

See also "upstream" in this *Glossary*.

drawings

Scale diagrams of the roadway in the vicinity of the work zone that identify the planned arrangement of traffic control devices in accordance with this Manual.

Drawings include dimensions and show all painted markings and physical features (signs, no-post guardrail, lamp standards, etc.) that may affect traffic operations, roadway geometry, and lane configurations.

drop-off

An abrupt change in the road level created by construction activities—such as milling, paving, or excavating—that is steeper than 3:1 (i.e., a non-traversable slope).

See also *Section 6.5: Treatment of Drop-Offs and Travel Lane Excavations.*

dynamic message sign

A programmable traffic control device that displays messages composed of letters, symbols/graphics, or both, and is used to provide drivers with highway condition information or to warn or manage traffic.

The acronym for dynamic message sign is DMS. It may also be called a changeable message sign (CMS) or a variable message sign (VMS).

See also *Section 4.3: Dynamic Message Signs.*

emergency

Situation which requires immediate response to save lives or prevent serious injury using whatever resources are available.

emergent work

Unplanned, quick-response work necessitated by an unanticipated situation that presents a risk to road users.

Each entry onto the travelled portion of the highway to perform emergent work takes less than 1 minute, and the total time required to complete the work is less than 5 minutes. (This does not apply for emergency incidents.)

escort vehicle	A vehicle used in quick-dry paint operations on two-lane, two-way roadways as a warning device to the travelling public. Typically driven ahead of the paint truck, it carries supplies, transports personnel between job sites, and communicates information about highway conditions, hazards, and traffic flow to other vehicles involved in the painting operation.
flagging	The control/direction of traffic using Automated Flagger Assistance Devices (AFADs) or Traffic Control Persons (TCPs).
flasher	A yellow flashing warning light.
freeway	A divided primary highway with two or more lanes in each direction and access via grade-separated interchanges only. Posted speed limit is typically 100 km/h or higher.
guidance	Uses words such as **should, recommended** - describes a recommended, but not mandatory, practice. Decisions contrary to guidance are documented. See *Section 6.11: Using Sections 7 to 19*.
high speed	A speed of 70 km/h or higher. See also “low speed” in this *Glossary*.
high-volume roadway	During work, a roadway which carries 1,000 or more vehicles per day. See also “low-volume roadway” in this *Glossary*.
highway	Every road, street, lane or right of way designed or intended for or used by the general public for the passage of traffic.
Implementation Plan	A sub-plan within a Traffic Management Plan that identifies the designated Traffic Control Manager and Traffic Control Supervisor and their qualifications, responsibilities, and duties, as well as procedures for ensuring that traffic management sub-plans are implemented in a coordinated manner (see *Section 3: Traffic Management Plans*).
inactive work site	A portion of roadway or right-of-way on which work has commenced but has temporarily ceased, and which has not been returned to normal operating conditions.
incident	In the context of Traffic Management, an incident refers to an event which affects Traffic Operations for workers and/or the travelling public, such as a collision which occurs within the construction zone.

Incident Management Plan	A sub-plan within a Traffic Management Plan that documents a plan for detecting incidents and managing incident response operations (see *Section 3: Traffic Management Plans*). It includes priorities and procedures for incident detection, response actions that will restore traffic flow as quickly as possible, and a review and analysis process for reducing incident frequency and severity.
intermittently-moving work	A road maintenance activity with a frequently changing work zone (e.g., some mowing operations) or one that involves frequent stops that last 30 minutes or less (temporary patching, group re-lamping of street lights, Benkleman beam testing, crack sealing, etc.).
isolated pothole patching	The patching of one or two potholes within a 1 km length of roadway. See also "multiple pothole patching" in this *Glossary*.
lane closure	One (or more than one) lane of traffic is closed to traffic use in one or both directions but the entire road has a minimum of one lane available for the passage of traffic for each direction.
Lane Closure Request Form	The Ministry's "Work Notification/Lane Closure Request and Approval Form" that is completed by the Prime Contractor and submitted to the District Manager of Transportation for acceptance. It identifies the proposed work zone location and traffic control measures, and is the minimum level of documentation required from the Prime Contractor. See *Appendix E: Lane Closure Request Form* for a sample of the form and a link to it.
lane drop	The closure of a through lane by using appropriate temporary traffic control devices, including flashing arrow boards and merge tapers.
layout	A schematic diagram of the roadway showing the placement and general arrangement of traffic control devices. See also *Section 3.2: Traffic Management Sub-Plans* and the traffic control layouts in *Sections 7 through 19*.
line-type utility vehicle	A vehicle carrying personnel who are working on utility lines, such as power, phone, or fibre optic lines.

long-duration work	Planned work that occupies one location for more than one daylight period. Night work lasting more than 15 minutes is also considered long-duration work. See also "short-duration work" in this *Glossary*.
low speed	A speed of 60 km/h or less. See also "high speed" in this *Glossary*.
low-volume roadway	During work, a roadway which carries fewer than 1,000 vehicles per day. See also "high-volume roadway" in this *Glossary*.
Manual	This Traffic Management Manual for Work on Roadways (TMM).
may	Describes a permissive condition. It is optional and carries no requirement or recommendation.
Ministry	The British Columbia Ministry of Transportation and Infrastructure, which is the provincial government entity responsible for work on Provincial roadways and right-of-way. When Ministry responsibilities are identified in this Manual, municipal Road Authorities may have similar responsibilities for work on municipal roads and right-of-way.
mobile work	Continuously slow-moving work or intermittently-moving work, with short stops of 30 minutes or less. The traffic control devices for mobile work are typically vehicle-mounted.
multilane divided roadway	A roadway with two or more travel lanes in each direction, including passing or climbing lanes, where the directions of travel are physically separated by a physical barrier.
multilane roadway	A roadway with two or more travel lanes in at least one direction, including climbing and passing lanes.
multiple pot-hole patching	The patching of multiple clusters of potholes along a stretch of roadway such that the work crew needs to stop several times in succession within 1 kilometre. It is a type of mobile work. See also "isolated pothole patching" in this Glossary.
near miss	In the context of Traffic Management, a near miss refers to an event where abrupt corrective action is required in order to avoid a collision. This unplanned event did not result in injury, illness, or damage, but had the potential to do so.

option	Uses words such as may - describes a permissive condition. It is optional and carries no requirement or recommendation. See *Section 6.11: Using Sections 7 to 19*
pilot car	For the purposes of this Manual, a vehicle marked with warning signs and lights that is used to guide a queue of vehicles through a work zone or detour (sometimes called "pilot vehicle"). (For links to information on piloting extraordinary loads, see the websites cited at the beginning of Section 4.11.9: Pilot Cars for Work Zones).
pilot vehicle operation delay	The amount of time elapsed when a vehicle joins the back of the queue until it passes by the traffic control device (e.g. traffic control person) to follow the pilot vehicle.
platoon	A group of vehicles or pedestrians travelling together, either voluntarily or involuntarily, because of traffic signal controls, other traffic control devices, road geometrics, or other factors.
portable signal	See "temporary traffic control signal" in this *Glossary*.
Prime Contractor	The organization directly constructing or maintaining works on a Provincial highway and responsible for: • obtaining Ministry authorization to carry out the works • developing an acceptable Traffic Management Plan • implementing the Plan in accordance with Ministry requirements
project	A work operation or activity undertaken on a roadway or right-of-way and requiring temporary traffic control.
project category	A project classification (Category 1, 2, or 3) based on the project's anticipated effect on traffic operations and the traffic control required for the works (see *Section 3: Traffic Management Plans*).
Public Information Plan	A sub-plan within a Traffic Management Plan that identifies actions and procedures for informing the travelling public, project stakeholders, and Ministry staff of current traffic operations and planned changes to traffic operations (see *Section 3: Traffic Management Plans*).
quality assurance	A process which ensures the Prime Contractor is following their Quality Control Plan.

quality control	An inspection process that examines the effectiveness of the temporary traffic control. It is the responsibility of the Prime Contractor.
Quality Control Plan	A plan which documents the Prime Contractor's quality control inspection process.
queue clearing time	The minimum amount of time that the highway must remain open to clear queued traffic and restore free-flow operation prior to implementing a subsequent Road Closure.
quick-dry pavement marking	A form of pavement marking in which the paint dries rapidly (typically in 90 seconds or less), and where paint is not tracked into the travel lanes by vehicles driving over the marking. See also "conventional pavement marking" in this Glossary and *Section 14.1: General Information - Pavement Marking*.
random minor traffic interruption(s)	A very brief stoppage in traffic in one or more directions for construction activities.
regular posted speed limit	The posted speed limit of the highway prior to any planned work. This is the speed limit the Ministry has established through the H-223 form, as signed by the Chief Engineer.
Road Authority	The jurisdiction that is responsible for operating the road. For Provincial jurisdictions, the Road Authority is typically the District Manager of Transportation or delegate.
roadside diversion	A deviation from the normal roadway where a section of the road is closed by road works and a short detour is therefore required, usually within the right-of-way, to bypass the work activity area.
roadway	The portion of a street or highway that is normally used for vehicular traffic. The roadway excludes the shoulder.
road closure(s)	A stoppage of traffic in one or both directions for the purpose of blasting rock, tie-ins, girder erection and paving activities, etc.
road users	Anyone who uses or crosses a road, including but not limited to vehicles, cyclists, pedestrians, and mobility devices. See also "traffic" in this *Glossary.*
roll-ahead buffer distance	The longitudinal distance measured from the front of the buffer vehicle to the work activity area. It provides a margin of safety in case of impact.

scheduled stoppage	A stoppage of traffic in one or both directions.
shadow vehicle	A vehicle used primarily in slow-moving operations as a mobile advance warning and sign support device. It may travel on the roadway or on the shoulder.
shall	Describes a mandatory condition. For provincial highways, if a mandatory condition cannot be met, contact Traffic and Highway Safety Engineering - Policy and Standards for direction.
short-duration work	Planned work which occupies one location for more than 15 minutes during a single daylight period. See also "long-duration work" in this *Glossary*.
should	Describes a recommended, but not mandatory, practice. Decisions contrary to a "should" are required to be documented. The documentation process is discussed in *Section 6.11.2: Application of Sections 7 to 19.* See also "guidance" in this *Glossary*.
shoulder	That part of a roadway contiguous with the travelled way intended for emergency stopping, and/or lateral support of the roadway structure. It may also be configured to be accessible for pedestrian and bicycle use. Typically the paved area outside of the highway's painted lane edge (fog lines). If not painted, the area outside the typical travel lane width of 3.5 to 3.7 metres.
simple project	A Category 1 or Category 2 project which may not require an Incident Management Plan, or a Public Information Plan, and for which no specific risk issues have been identified (see *Section 3: Traffic Management Plans*).
single lane alternating traffic (SLAT)	A traffic control practice typically used on a two-lane, two-way roadway whereby one direction of traffic is held while the other is permitted to proceed, and then vice versa. This process is repeated successively so that traffic continues to flow with minimal delays. It is a method acceptable to Road Authorities because delays are typically less than 5 minutes.